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Abstract. abbbaaaaaaaa

1. Introduction

2. Model

3. Introduction and model formulation

4. Model formulation

Barkley's reduced description of transitional pipe flow couples turbulence intensity to the local centerline velocity [1, 3, 2]. Suppressing the spatial-derivative terms gives the homogeneous reaction system

$$\begin{aligned} \dot{q} &= q\left[r + u - 2 - (r+\delta)(q-1)^2\right], \\ \dot{u} &= \varepsilon\left[2 - u + \kappa q(1-u)\right], \end{aligned} \qquad r, \delta, \varepsilon, \kappa > 0. \tag{1}$$

Here $q \geq 0$ measures turbulence intensity, u denotes the normalized centerline velocity, and r is a Reynolds-number-related control parameter. The velocity normalization assigns the value $u = 2$ to the laminar state, while the turbulence-dependent term drives u toward the reference level $u = 1$. The parameter ε controls the relative timescale of velocity relaxation, and κ measures the strength of its coupling to turbulence.

The parameter δ controls the linear decay of turbulence near the laminar equilibrium $(q, u) = (0, 2)$: the corresponding eigenvalues are $-\delta$ and $-\varepsilon$. Thus $\delta > 0$ preserves the local stability of the laminar state, while $\kappa > 0$ represents the reduction of centerline velocity associated with increasing turbulence.

In this investigation, we adopt the parameter choices $\delta = 0.1$ and $\kappa = 2$, obtaining

$$\begin{aligned} \dot{q} &= q\big[r + u - 2 - (r+0.1)(q-1)^2\big], \\ \dot{u} &= \varepsilon\big[2 - u + 2q(1-u)\big], \end{aligned} \qquad r, \varepsilon > 0. \tag{2}$$

These choices fix the laminar decay rate and the turbulence–velocity coupling strength, leaving r and ε as the bifurcation parameters. System (2) is a cubic polynomial differential system. Our aim is to investigate the existence, stability, and number of its limit cycles in the physical half-plane $q \geq 0$. Such cycles describe spatially homogeneous temporal oscillations of the reduced model; they should be distinguished from traveling or spatially localized structures of the full pipe-flow equations.

5. Equilibria and existence of limit cycles

The onset of a periodic orbit depends on the geometry and stability of the underlying equilibria. We therefore first locate the positive equilibrium branches and classify their stability throughout the parameter plane. We then identify the Hopf curve and determine the stability of the periodic orbit born from it. Throughout this section, we consider

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system (2) with $r, \varepsilon > 0$ and write $A = r + 1/10$. The parameters of the model remain r and ε; equilibrium coordinates introduced below are determined by r.

5.1. **The physical region and the equilibrium branches.** The line $q = 0$ is invariant, and the equation on this line is $\dot{u} = \varepsilon(2 - u)$. Hence a nonconstant periodic orbit in $q \geq 0$ must lie in $q > 0$. The strip

$$\mathcal{S} = \{(q, u) : q \geq 0,\ 1 \leq u \leq 2\}$$

is positively invariant: on $u = 1$ one has $\dot{u} = \varepsilon > 0$, whereas on $u = 2$ one has $\dot{u} = -2\varepsilon q \leq 0$. In fact, every nonconstant periodic orbit in the physical half-plane is contained in

$$\Omega = \{(q, u) : 0 < q < 2,\ 1 < u < 2\}. \tag{3}$$

Indeed, at an extremum of u along such an orbit,

$$u = 1 + \frac{1}{1 + 2q} \in (1, 2).$$

At a maximum of q, the first equation gives

$$A(q - 1)^2 = r + u - 2 < r < A,$$

which implies $q < 2$.

The laminar equilibrium is $E_0 = (0, 2)$. Every positive equilibrium has the form

$$E_* = \left(q_*, 1 + \frac{1}{1 + 2q_*}\right), \qquad P_r(q_*) = 0, \tag{4}$$

where

$$P_r(q) = 2Aq^3 - 3Aq^2 + 2(1 - r)q + \frac{1}{10}. \tag{5}$$

All positive equilibria satisfy $0 < q_* < 2$. Solving the equilibrium relation for r gives

$$\mathcal{R}(q) = \frac{2q + \frac{1}{10}(1 + 2q)(q - 1)^2}{(1 + 2q)q(2 - q)}, \qquad 0 < q < 2. \tag{6}$$

Its derivative is

$$\mathcal{R}'(q) = \frac{44q^3 - 30q^2 - 3q - 1}{5q^2(2 - q)^2(1 + 2q)^2}. \tag{7}$$

The numerator has exactly one positive zero, denoted by q_{SN}. Since $\mathcal{R}(q) \to +\infty$ as $q \to 0^+$ or $q \to 2^-$, this zero is the unique minimum point of $\mathcal{R}$. Define

$$\begin{aligned} &44q_{\mathrm{SN}}^3 - 30q_{\mathrm{SN}}^2 - 3q_{\mathrm{SN}} - 1 = 0, \qquad r_{\mathrm{SN}} = \mathcal{R}(q_{\mathrm{SN}}), \\ &q_{\mathrm{SN}} \simeq 0.802140181960, \qquad r_{\mathrm{SN}} \simeq 0.645189834506. \end{aligned} \tag{8}$$

Equivalently, r_{SN} is the unique real root of

$$2500r^3 - 5700r^2 + 5061r - 1564 = 0.$$

Thus the positive equilibria are classified as follows:

(1) If $0 < r < r_{\mathrm{SN}}$, there is no positive equilibrium.
(2) If $r = r_{\mathrm{SN}}$, the two positive branches coalesce at

$$E_{\mathrm{SN}} = \left(q_{\mathrm{SN}}, 1 + \frac{1}{1 + 2q_{\mathrm{SN}}}\right).$$

(3) If $r > r_{\mathrm{SN}}$, there are exactly two positive equilibria

$$\begin{aligned} E_\pm(r) &= \left(q_\pm(r), 1 + \frac{1}{1 + 2q_\pm(r)}\right), \\ 0 < q_-(r) &< q_{\mathrm{SN}} < q_+(r) < 2. \end{aligned} \tag{9}$$

Here q_* denotes a general positive equilibrium coordinate, whereas q_{SN} is the fixed coordinate at which the branches merge. In the Hopf calculation below, $q_* = q_+(r)$.

5.2. **Stability and the node–focus boundaries.** At the laminar equilibrium,

$$J(E_0) = \begin{pmatrix} -1/10 & 0 \\ -2\varepsilon & -\varepsilon \end{pmatrix}.$$

Consequently, E_0 is an asymptotically stable node for every $r, \varepsilon > 0$. Its eigenvalues are distinct if $\varepsilon \neq 1/10$. At $\varepsilon = 1/10$, it is an improper stable node with a repeated eigenvalue and one eigendirection.

At a positive equilibrium, put $D_* = 1 + 2q_*$. The Jacobian is

$$J(E_*) = \begin{pmatrix} 2Aq_*(1-q_*) & q_* \\ -2\varepsilon/D_* & -\varepsilon D_* \end{pmatrix}. \tag{10}$$

The eigenvalues of $J(E_*)$ satisfy

$$\lambda^2 - T_*\lambda + \Delta_* = 0,$$

where $T_* = \operatorname{tr} J(E_*)$ and $\Delta_* = \det J(E_*)$. Direct calculation gives

$$\begin{aligned} T_* &= 2Aq_*(1-q_*) - \varepsilon D_*, \\ \Delta_* &= \varepsilon\left[\frac{2q_*}{D_*} - 2Aq_*(1-q_*)D_*\right] = \varepsilon q_* P_r'(q_*), \qquad D_* = 1 + 2q_*. \end{aligned} \tag{11}$$

The last equality uses the equilibrium relation $P_r(q_*) = 0$. Indeed, direct expansion yields

$$\begin{aligned} &\frac{2q}{1+2q} - 2Aq(1-q)(1+2q) \\ &\qquad = qP_r'(q) - \frac{2q}{1+2q}P_r(q). \end{aligned}$$

Evaluating this identity at $q = q_*$ proves the determinant formula in (11).

We first use the determinant to distinguish the two positive equilibrium branches. From

$$P_r(q) = (1+2q)q(2-q)\big(\mathcal{R}(q) - r\big), \qquad 0 < q < 2,$$

differentiation at a positive equilibrium gives

$$P_r'(q_*) = (1+2q_*)q_*(2-q_*)\mathcal{R}'(q_*).$$

The factor multiplying $\mathcal{R}'(q_*)$ is positive. Since $\mathcal{R}$ is decreasing on the lower branch and increasing on the upper branch, it follows that

$$P_r'(q_-) < 0, \qquad P_r'(q_+) > 0.$$

Consequently,

$$\Delta_- < 0, \qquad \Delta_+ > 0.$$

Thus E_- always has two real eigenvalues of opposite signs and is a hyperbolic saddle.

For the upper equilibrium, the positive determinant implies that stability is determined by the trace whenever the trace is nonzero. More precisely, E_+ is asymptotically stable when $T_+ < 0$ and unstable when $T_+ > 0$. The case $T_+ = 0$ must be examined separately because the eigenvalues then have zero real part.

Fix $r > r_{\mathrm{SN}}$ and henceforth write

$$q_* = q_+(r), \qquad D = 1 + 2q_*, \qquad m = 2Aq_*(1-q_*).$$

It is also convenient to introduce the positive quantity

$$d = \frac{2q_*}{D} - mD = q_* P_r'(q_*) > 0.$$

These quantities depend on r and are independent of ε. The trace and determinant now take the simple form

$$T_+ = m - \varepsilon D, \qquad \Delta_+ = \varepsilon d.$$

The sign of m is determined by the position of q_* relative to 1. Since $\mathcal{R}$ is strictly increasing on the upper branch and $\mathcal{R}(1) = 2/3$, there are three cases.

(1) Suppose $r_{\mathrm{SN}} < r < 2/3$. Then $q_* < 1$ and $m > 0$. There is exactly one positive value of ε at which the trace vanishes:

$$\varepsilon_H(r) = \frac{m}{D} = \frac{2(r + 1/10)q_+(r)(1 - q_+(r))}{1 + 2q_+(r)}. \tag{12}$$

Indeed,

$$T_+ = D\big(\varepsilon_H(r) - \varepsilon\big).$$

It follows that E_+ is unstable for $0 < \varepsilon < \varepsilon_H(r)$ and asymptotically stable for $\varepsilon > \varepsilon_H(r)$.

At $\varepsilon = \varepsilon_H(r)$,

$$T_+ = 0, \qquad \Delta_+ = \varepsilon_H(r)d > 0,$$

so the eigenvalues are

$$\lambda_{1,2} = \pm i\sqrt{\varepsilon_H(r)d}.$$

This identifies a Hopf candidate. The nonlinear nondegeneracy and the stability at the critical parameter are established by the Lyapunov-coefficient calculation below.

(2) Suppose $r = 2/3$. Then $q_* = 1$ and $m = 0$, giving

$$T_+ = -3\varepsilon < 0, \qquad \Delta_+ = \frac{2}{3}\varepsilon > 0.$$

Hence E_+ is asymptotically stable for every $\varepsilon > 0$.

(3) Suppose $r > 2/3$. Then $q_* > 1$ and $m < 0$. Therefore

$$T_+ = m - \varepsilon D < 0$$

for every $\varepsilon > 0$, while $\Delta_+ > 0$. Thus E_+ is again asymptotically stable for every $\varepsilon > 0$.

To complete the linear classification, we distinguish nodes from foci. The eigenvalues at E_+ can be written as

$$\lambda_{1,2} = \frac{T_+ \pm \sqrt{\mathcal{D}(\varepsilon)}}{2},$$

where

$$\mathcal{D}(\varepsilon) = T_+^2 - 4\Delta_+$$

is the characteristic discriminant. Substituting the expressions for T_+ and Δ_+ gives

$$\begin{aligned}\mathcal{D}(\varepsilon) &= (m - \varepsilon D)^2 - 4\varepsilon d \\ &= D^2\varepsilon^2 - 2(mD + 2d)\varepsilon + m^2.\end{aligned}$$

Thus the node–focus boundaries are obtained by solving a quadratic equation in ε.

To express its roots more clearly, set

$$b = \frac{2q_*}{D} > 0.$$

Since $d = b - mD$, one has

$$mD + 2d = b + d.$$

The two roots are consequently

$$\begin{aligned}\varepsilon_{\mathrm{nf}}^{\pm}(r) &= \frac{b + d \pm 2\sqrt{bd}}{D^2} \\ &= \frac{(\sqrt{b} \pm \sqrt{d})^2}{D^2}.\end{aligned} \tag{13}$$

Both roots are nonnegative, and they are distinct because $b, d > 0$. Moreover, $\varepsilon_{\mathrm{nf}}^{-} = 0$ holds precisely when $b = d$, or equivalently when $m = 0$.

The discriminant therefore factors as

$$\mathcal{D}(\varepsilon) = D^2\big(\varepsilon - \varepsilon_{\mathrm{nf}}^{-}\big)\big(\varepsilon - \varepsilon_{\mathrm{nf}}^{+}\big). \tag{14}$$

Since the coefficient D^2 is positive, this factorization has the following interpretation:

(1) For $0 < \varepsilon < \varepsilon_{\mathrm{nf}}^{-}$ or $\varepsilon > \varepsilon_{\mathrm{nf}}^{+}$, the discriminant is positive. The eigenvalues are real and have the same sign, so E_+ is a node. Its stability is determined by the sign of T_+.

(2) For $\varepsilon_{\mathrm{nf}}^{-} < \varepsilon < \varepsilon_{\mathrm{nf}}^{+}$, the discriminant is negative. The eigenvalues are complex conjugates with real part $T_+/2$. Hence E_+ is a stable focus when $T_+ < 0$ and an unstable focus when $T_+ > 0$. If $T_+ = 0$, the eigenvalues are purely imaginary and the linearization alone does not determine stability.

(3) At a positive threshold $\varepsilon = \varepsilon_{\mathrm{nf}}^{\pm}$, the eigenvalues coincide. Since

$$T_+^2 = 4\Delta_+ > 0,$$

the repeated eigenvalue $T_+/2$ is nonzero. Furthermore, the off-diagonal entry $q_* > 0$ prevents $J(E_+)$ from being a scalar matrix. There is therefore only one eigendirection, and E_+ is an improper node. It is stable if $T_+ < 0$ and unstable if $T_+ > 0$.

In particular, the node–focus boundaries preserve hyperbolicity and do not represent a loss of equilibrium stability.

For $r_{\mathrm{SN}} < r < 2/3$, the Hopf threshold lies strictly between the two node–focus thresholds. Indeed, $m > 0$ implies $b > d > 0$, so $\varepsilon_{\mathrm{nf}}^{-} > 0$, while

$$\mathcal{D}(\varepsilon_H) = -4\varepsilon_H d < 0.$$

The factorization (14) then yields

$$0 < \varepsilon_{\mathrm{nf}}^{-}(r) < \varepsilon_H(r) < \varepsilon_{\mathrm{nf}}^{+}(r). \tag{15}$$

Thus increasing ε first changes the unstable node into an unstable focus, then crosses the trace-zero threshold, and finally changes the stable focus into a stable node.

At $r = 2/3$, one has $D = 3$ and $b = d = 2/3$, giving

$$\varepsilon_{\mathrm{nf}}^{-} = 0, \qquad \varepsilon_{\mathrm{nf}}^{+} = \frac{8}{27}.$$

Accordingly, E_+ is a stable focus for $0 < \varepsilon < 8/27$, an improper stable node at $\varepsilon = 8/27$, and a stable node for $\varepsilon > 8/27$.

For $r > 2/3$, the trace remains negative throughout the positive ε-axis. Both node–focus thresholds are positive: E_+ is a stable focus between them, a stable node outside them, and an improper stable node at either threshold. In particular, no Hopf bifurcation of E_+ occurs for $r \geq 2/3$.

The complete classification of E_+ is given in Table 1. The nonlinear classification at $\varepsilon = \varepsilon_H$ is established in Subsection 5.4.

5.3. **The coalescence point and its double-zero exception.** At $r = r_{\mathrm{SN}}$, write

$$s = q_{\mathrm{SN}}, \qquad A_{\mathrm{SN}} = r_{\mathrm{SN}} + \frac{1}{10}, \qquad D_{\mathrm{SN}} = 1 + 2s.$$

Define

$$\varepsilon_{\mathrm{BT}} = \frac{2A_{\mathrm{SN}}s(1-s)}{D_{\mathrm{SN}}} \simeq 0.090827436864. \tag{16}$$

The eigenvalues at E_{SN} are 0 and $D_{\mathrm{SN}}\big(\varepsilon_{\mathrm{BT}} - \varepsilon\big)$.

TABLE 1. Classification of the upper positive equilibrium E_+. The equilibrium E_0 is always an asymptotically stable node, and E_- is always a saddle when the two positive branches are distinct. All thresholds in a row are evaluated at the same fixed r.

r-range	ε-range	Type of E_+
$r_{\mathrm{SN}} < r < 2/3$	$(0, \varepsilon_{\mathrm{nf}}^-)$	Unstable node
	$\varepsilon_{\mathrm{nf}}^-$	Unstable improper node
	$(\varepsilon_{\mathrm{nf}}^-, \varepsilon_H)$	Unstable focus
	ε_H	Unstable weak focus
	$(\varepsilon_H, \varepsilon_{\mathrm{nf}}^+)$	Stable focus
	$\varepsilon_{\mathrm{nf}}^+$	Stable improper node
	$(\varepsilon_{\mathrm{nf}}^+, \infty)$	Stable node
$r = 2/3$	$(0, 8/27)$	Stable focus
	$8/27$	Stable improper node
	$(8/27, \infty)$	Stable node
$r > 2/3$	$(0, \varepsilon_{\mathrm{nf}}^-) \cup (\varepsilon_{\mathrm{nf}}^+, \infty)$	Stable node
	$\varepsilon_{\mathrm{nf}}^-,\ \varepsilon_{\mathrm{nf}}^+$	Stable improper node
	$(\varepsilon_{\mathrm{nf}}^-, \varepsilon_{\mathrm{nf}}^+)$	Stable focus

For $\varepsilon \neq \varepsilon_{\mathrm{BT}}$, this is a nondegenerate saddle-node. Its transverse direction is repelling when $0 < \varepsilon < \varepsilon_{\mathrm{BT}}$ and attracting when $\varepsilon > \varepsilon_{\mathrm{BT}}$. The fold is nondegenerate because

$$P''_{r_{\mathrm{SN}}}(s) = 6A_{\mathrm{SN}}(2s-1) > 0,$$

and

$$\left.\frac{\partial P_r(s)}{\partial r}\right|_{r=r_{\mathrm{SN}}} = s(2s+1)(s-2) < 0.$$

At $\varepsilon = \varepsilon_{\mathrm{BT}}$, both eigenvalues vanish and the Jacobian is nonzero and nilpotent. The nonlinear terms identify this exception as a nondegenerate Bogdanov–Takens point. To verify this, introduce the local coordinate $v = \dot{q}$. The map $(q, u) \mapsto (q, v)$ is invertible for $q > 0$ because $\partial v/\partial u = q$. In these coordinates,

$$\dot{q} = v, \qquad \dot{v} = \mathcal{F}(q, v; r, \varepsilon), \tag{17}$$

with

$$\mathcal{F}(q, v; r, \varepsilon) = -\varepsilon q P_r(q) + \big[2Aq(1-q) - \varepsilon(1+2q)\big]v + \frac{v^2}{q}. \tag{18}$$

At the double-zero point, setting $x = q - s$ gives

$$\dot{x} = v, \qquad \dot{v} = a_{20}x^2 + a_{11}xv + s^{-1}v^2 + O\big(\|(x,v)\|^3\big),$$

where

$$\begin{aligned} a_{20} &= -3\varepsilon_{\mathrm{BT}}A_{\mathrm{SN}}s(2s-1) < 0, \\ a_{11} &= 2A_{\mathrm{SN}}(1-2s) - 2\varepsilon_{\mathrm{BT}} < 0. \end{aligned} \tag{19}$$

In particular, $a_{20}a_{11} \neq 0$.

The unfolding is regular as well. If

$$c_r = \varepsilon_{\mathrm{BT}}s^2 D_{\mathrm{SN}}(2-s) > 0,$$

then direct differentiation gives

$$\det D_{(q,v,r,\varepsilon)}\big(v, \mathcal{F}, \mathcal{F}_v, -\mathcal{F}_q\big)\big|_{\mathrm{BT}} = -2a_{20}c_r D_{\mathrm{SN}} \neq 0. \tag{20}$$

These are the standard nondegeneracy and regularity conditions for a Bogdanov–Takens bifurcation.

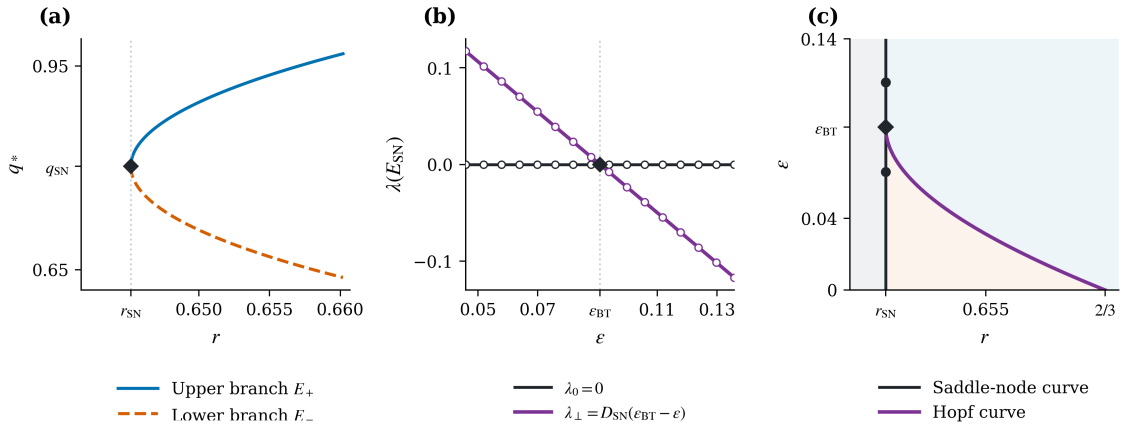


FIGURE 1. Equilibrium coalescence and the double-zero degeneracy. (a) The positive equilibrium branches E_- and E_+ merge at (r_{SN}, q_{SN}), marked by a black diamond. The lower branch consists of saddles, whereas the upper branch consists of nonsaddle equilibria. (b) Spectrum at the coalesced equilibrium along $r = r_{SN}$. One eigenvalue remains zero, while the transverse eigenvalue $\lambda_\perp = (1 + 2q_{SN})(\varepsilon_{BT} - \varepsilon)$ changes sign at $\varepsilon = \varepsilon_{BT}$. Lines represent the analytical expressions, and open circles show eigenvalues computed from the original Jacobian. (c) The saddle-node curve (black) and the Hopf curve (purple) meet at the Bogdanov–Takens point. Blue and orange shading indicate, respectively, asymptotic stability and instability of E_+; the gray region contains no positive equilibrium. The laminar equilibrium remains asymptotically stable throughout. The circles on the saddle-node curve identify the parameters used in panels (a) and (c) of Fig. 2. Here $r_{SN} \simeq 0.645189834506$, $q_{SN} \simeq 0.802140181960$, and $\varepsilon_{BT} \simeq 0.090827436864$.

5.4. **The Hopf curve and the emerging limit cycle.** Neither the laminar equilibrium nor the lower positive branch can undergo a Hopf bifurcation. For the upper branch, the trace can vanish at a positive value of ε precisely when $r_{SN} < r < 2/3$. Thus (12) gives the entire Hopf candidate curve in the physical half-plane. On this curve,

$$\lambda_{1,2} = \pm i\omega_H, \qquad \omega_H^2 = \varepsilon_H d = \varepsilon_H q_* P_r'(q_*) > 0. \tag{21}$$

The equilibria do not depend on ε, so the crossing condition is particularly simple:

$$\left.\frac{\partial \operatorname{Re}\lambda_{1,2}}{\partial \varepsilon}\right|_{\varepsilon=\varepsilon_H} = -\frac{D}{2} < 0. \tag{22}$$

The curve is strictly decreasing. Indeed, along the upper branch,

$$\varepsilon_H = \frac{(22q_* + 1)(1 - q_*)}{5(2 - q_*)(1 + 2q_*)^2},$$

and differentiation with respect to q_* gives

$$-\frac{44q_*^3 - 106q_*^2 + 166q_* - 35}{5(2 - q_*)^2(1 + 2q_*)^3} < 0, \qquad q_{SN} < q_* < 1.$$

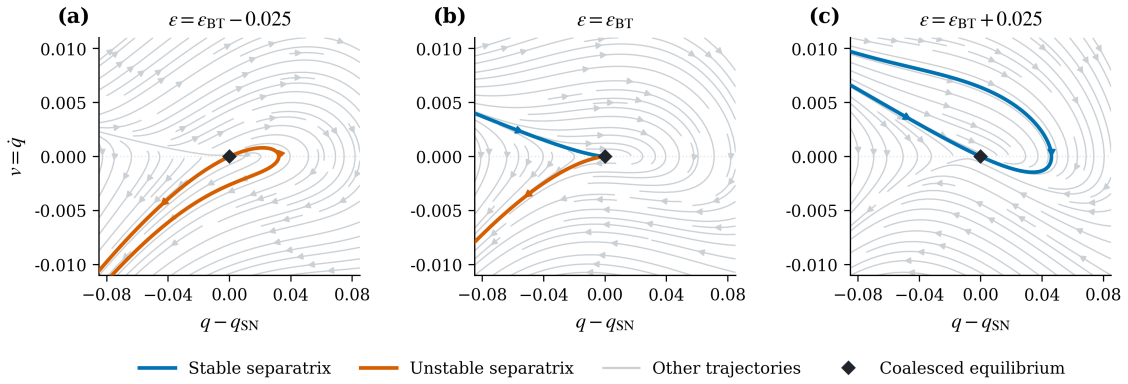


FIGURE 2. Local flow at the equilibrium coalescence $r = r_{SN}$, represented in the coordinates $(q - q_{SN}, v)$ with $v = \dot{q}$. The black diamond marks the coalesced equilibrium. (a) For $\varepsilon = \varepsilon_{BT} - 0.025$, the transverse eigenvalue is positive, and the orange curves highlight the strong unstable manifold of the saddle-node. (b) At $\varepsilon = \varepsilon_{BT}$, both eigenvalues vanish. The incoming blue branch and outgoing orange branch form the cusp separatrices of the nondegenerate Bogdanov–Takens equilibrium. (c) For $\varepsilon = \varepsilon_{BT} + 0.025$, the transverse eigenvalue is negative, and the blue curves highlight the strong stable manifold of the saddle-node. Gray streamlines represent nearby trajectories, and arrows indicate the forward-time orientation. Identical axis limits are used in all panels to make the change in local geometry directly comparable.

Since $q_+(r)$ is increasing, it follows that

$$\begin{gathered} \varepsilon_H'(r) < 0, \\ \lim_{r \downarrow r_{SN}} \varepsilon_H(r) = \varepsilon_{BT}, \qquad \lim_{r \uparrow 2/3} \varepsilon_H(r) = 0. \end{gathered} \tag{23}$$

Consequently, for each fixed $0 < \varepsilon < \varepsilon_{BT}$, there is exactly one $r_H(\varepsilon) \in (r_{SN}, 2/3)$ satisfying

$$\varepsilon = \varepsilon_H(r_H(\varepsilon)).$$

The upper equilibrium is unstable for $r_{SN} < r < r_H(\varepsilon)$ and asymptotically stable for $r > r_H(\varepsilon)$. If $\varepsilon \geq \varepsilon_{BT}$, it is asymptotically stable for every $r > r_{SN}$.

It remains to determine whether the imaginary eigenvalues produce a nondegenerate Hopf bifurcation and, if so, on which side the periodic orbit exists. For this purpose, set

$$K = 1 - A(1 - q_*)D^2.$$

Along the Hopf curve,

$$d = \frac{2q_*}{D} K > 0,$$

so $K > 0$. Translate the equilibrium and normalize the linear part by

$$X = q - q_*, \qquad Y = -\frac{m(q - q_*) + q_*(u - u_*)}{\omega_H}. \tag{24}$$

At the critical parameter, the resulting equations are

$$\begin{aligned} \dot{X} &= -\omega_H Y - A q_* X^2 - \frac{\omega_H}{q_*} XY - AX^3, \\ \dot{Y} &= \omega_H X + \frac{3mAq_*(2q_* - 1)}{D\omega_H} X^2 + \frac{m}{q_* D} XY + \frac{mA}{\omega_H} X^3. \end{aligned} \tag{25}$$

The standard planar Hopf calculation [**?**] then gives the cubic radial coefficient

$$c_H = \frac{A}{8K}\left[A\big(4q_*^3 + 6q_*^2 - 6q_* + 5\big) - 2\right]. \tag{26}$$

The numerical value of a Lyapunov coefficient depends on the normalization; the sign of c_H is the sign of the first Lyapunov coefficient.

This sign is strictly positive throughout the curve. Indeed, $q_* > q_{\mathrm{SN}} > 4/5$ and $A > 7/10$. The polynomial $4q^3 + 6q^2 - 6q + 5$ is increasing for $q \geq 4/5$, and its value at $4/5$ is $761/125 > 6$. Therefore the bracket in (26) is positive, and

$$c_H > 0 \qquad \text{for } r_{\mathrm{SN}} < r < 2/3. \tag{27}$$

Every Hopf bifurcation on this curve is consequently nondegenerate and subcritical. In particular, no generalized Hopf point occurs on the curve.

For a fixed r in this interval, put $\mu = \varepsilon - \varepsilon_H(r)$. In a local radial coordinate, the normal form can be written as

$$\dot\rho = -\frac{D}{2}\mu\rho + c_H\rho^3 + O\big(\rho^5 + |\mu|\rho^3 + \mu^2\rho\big). \tag{28}$$

Thus a unique sufficiently small limit cycle exists for $\mu > 0$ sufficiently small, and its leading amplitude satisfies

$$\rho^2 = \frac{D}{2c_H}\big(\varepsilon - \varepsilon_H(r)\big) + O\big((\varepsilon - \varepsilon_H(r))^2\big). \tag{29}$$

The cycle is hyperbolic and unstable. It surrounds the asymptotically stable equilibrium E_+ and shrinks to it as $\varepsilon \downarrow \varepsilon_H(r)$. At the critical parameter, E_+ is an unstable weak focus. For $\mu < 0$ sufficiently small, no sufficiently small limit cycle surrounds E_+. Equivalently, with $0 < \varepsilon < \varepsilon_{\mathrm{BT}}$ fixed, the small unstable cycle exists for $r > r_H(\varepsilon)$ sufficiently close to $r_H(\varepsilon)$.

The two endpoints of the Hopf curve have a different character. The endpoint $(r_{\mathrm{SN}}, \varepsilon_{\mathrm{BT}})$ is the double-zero point described above, whereas $(2/3, 0)$ lies on the excluded boundary $\varepsilon = 0$. Neither endpoint is an ordinary Hopf bifurcation.

5.5. **Two global nonexistence regions.** The local existence result can be complemented by two explicit regions in which no periodic orbit occurs anywhere in $q \geq 0$.

First, suppose $0 < r \leq r_{\mathrm{SN}}$. The equilibrium classification implies

$$P_r(q) > 0 \qquad (q > 0)$$

when $r < r_{\mathrm{SN}}$, and $P_{r_{\mathrm{SN}}}(q) \geq 0$, with equality only at $q = q_{\mathrm{SN}}$, when $r = r_{\mathrm{SN}}$. On a hypothetical nonconstant periodic orbit, let q attain its minimum. There $\dot q = 0$, and the equations give

$$\ddot q = q\dot u = -\varepsilon q P_r(q). \tag{30}$$

This is negative unless $r = r_{\mathrm{SN}}$ and $q = q_{\mathrm{SN}}$. In the exceptional case, $\dot q = \dot u = 0$, so the point is an equilibrium and cannot lie on a nonconstant periodic orbit. Both alternatives contradict the existence of a minimum on such an orbit.

Second, in the simply connected domain $q > 0$, use the Dulac function $B(q, u) = 1/q$. Writing the two components of the original vector field as F and G, one obtains

$$\frac{\partial(BF)}{\partial q} + \frac{\partial(BG)}{\partial u} = \frac{2Aq(1-q) - \varepsilon(1+2q)}{q}. \tag{31}$$

Let

$$\varepsilon_D(r) = (2 - \sqrt{3})\left(r + \frac{1}{10}\right), \qquad q_D = \frac{\sqrt{3} - 1}{2}. \tag{32}$$

The numerator in (31) becomes

$$-2A(q - q_D)^2 - \big(\varepsilon - \varepsilon_D(r)\big)(1 + 2q).$$

It is nonpositive for $\varepsilon \geq \varepsilon_D(r)$ and does not vanish identically on any open set. The Bendixson–Dulac criterion therefore excludes periodic orbits, including at equality. Hence every periodic orbit in the physical half-plane must satisfy the necessary parameter restrictions

$$\boxed{r > r_{\mathrm{SN}}, \qquad 0 < \varepsilon < \varepsilon_D(r).} \tag{33}$$

The equilibrium classification and the periodic-orbit results are collected in the following theorem.

Theorem 5.1 (Equilibria and the local birth of a limit cycle)**.** *Consider system* (2) *with* $r, \varepsilon > 0$ *in* $q \geq 0$*, with the equilibrium branches and thresholds defined above. Then:*

(1) The laminar equilibrium $E_0 = (0, 2)$ *is always an asymptotically stable node. There are no positive equilibria for* $r < r_{\mathrm{SN}}$*, one coalesced positive equilibrium for* $r = r_{\mathrm{SN}}$*, and exactly two distinct positive equilibria* E_- *and* E_+ *for* $r > r_{\mathrm{SN}}$*.*

(2) Whenever the positive branches are distinct, E_- *is a hyperbolic saddle. For* $r_{\mathrm{SN}} < r < 2/3$*, the equilibrium* E_+ *is unstable if* $0 < \varepsilon < \varepsilon_H(r)$ *and asymptotically stable if* $\varepsilon > \varepsilon_H(r)$*. For* $r \geq 2/3$*, it is asymptotically stable for every* $\varepsilon > 0$*. Its node–focus classification is given in Table 1.*

(3) At $r = r_{\mathrm{SN}}$ *and* $\varepsilon \neq \varepsilon_{\mathrm{BT}}$*, the positive equilibrium is a nondegenerate saddle-node, with a repelling transverse direction for* $\varepsilon < \varepsilon_{\mathrm{BT}}$ *and an attracting transverse direction for* $\varepsilon > \varepsilon_{\mathrm{BT}}$*. At*

$$(r, \varepsilon) = (r_{\mathrm{SN}}, \varepsilon_{\mathrm{BT}}),$$

it is a nondegenerate Bogdanov–Takens point.

(4) The only Hopf bifurcations in $q \geq 0$ *occur at* E_+ *on the curve*

$$r_{\mathrm{SN}} < r < 2/3, \qquad \varepsilon = \varepsilon_H(r).$$

This curve decreases strictly from $\varepsilon_{\mathrm{BT}}$ *to* 0*. Every point on it is a nondegenerate subcritical Hopf point, and* E_+ *is an unstable weak focus at the critical parameter. There is no generalized Hopf point on this curve.*

(5) For each fixed $r \in (r_{\mathrm{SN}}, 2/3)$*, there exist* $\eta(r) > 0$ *and a neighborhood* $U_r \subset \Omega$ *of* $E_+(r)$ *such that, for every*

$$\varepsilon_H(r) < \varepsilon < \varepsilon_H(r) + \eta(r),$$

there is exactly one limit cycle contained in U_r*. It is hyperbolic and unstable, and converges to* $E_+(r)$ *as* $\varepsilon \downarrow \varepsilon_H(r)$*. There is no limit cycle contained in* U_r *for*

$$\varepsilon_H(r) - \eta(r) < \varepsilon \leq \varepsilon_H(r).$$

(6) No nonconstant periodic orbit exists in $q \geq 0$ *if* $0 < r \leq r_{\mathrm{SN}}$ *or if* $\varepsilon \geq \varepsilon_D(r)$*.*

Proof. The equilibrium count follows from the unique minimum of $\mathcal{R}$. The trace, determinant, and characteristic discriminant in (11) and (14) give the equilibrium classification. The fold and double-zero assertions follow from the nondegeneracy calculations in Subsection 5.3. The crossing condition (22) and the positivity of the cubic coefficient (27) yield the Hopf assertions and the local cycle count. Finally, the minimum argument and the Dulac calculation in Subsection 5.5 prove the global nonexistence statements. □

The uniqueness assertion in Theorem 5.1 is local to the Hopf equilibrium. The restrictions (33) are global necessary conditions. Determining the maximal continuation of the periodic branch and excluding additional cycles outside the Hopf neighborhood require a separate global analysis.

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The nonzero equilibria satisfy

$$2(r + 0.1)q^3 - 3(r + 0.1)q^2 + 2(1 - r)q + 0.1 = 0.$$

Dividing by $2(r+0.1)$ and substituting $q = z + \frac{1}{2}$ reduces this algebraic equation to

$$z^3 + p(r)z + b(r) = 0,$$

where

$$p(r) = \frac{37 - 70r}{40(r+0.1)}, \qquad b(r) = \frac{21 - 30r}{40(r+0.1)}.$$

This substitution is used only to solve the equilibrium equation; it does not change the variables of the differential system.

For $r > r_c \approx 0.645189835$, define

$$s(r) = \sqrt{\frac{70r - 37}{120(r+0.1)}}, \qquad \theta(r) = \frac{1}{3}\arccos\left(\frac{30r - 21}{80(r+0.1)s(r)^3}\right). \tag{34}$$

Here arccos takes its principal value in $[0, \pi]$. The two positive roots are explicitly

$$\boxed{\begin{aligned} q_-(r) &= \frac{1}{2} + 2s(r)\cos\left(\theta(r) - \frac{2\pi}{3}\right), \\ q_+(r) &= \frac{1}{2} + 2s(r)\cos\theta(r). \end{aligned}} \tag{35}$$

They satisfy

$$0 < q_-(r) < q_+(r) < 2.$$

Consequently, the two positive equilibria are

$$\boxed{E_\pm(r) = \left(q_\pm(r), \frac{2[1 + q_\pm(r)]}{1 + 2q_\pm(r)}\right).} \tag{36}$$

The remaining root is

$$q_{\mathrm{n}}(r) = \frac{1}{2} + 2s(r)\cos\left(\theta(r) + \frac{2\pi}{3}\right),$$

and satisfies $-\frac{1}{2} < q_{\mathrm{n}}(r) < 0$. Its equilibrium is outside the physical half-plane.

At $r = r_c$, the same formulas hold by continuity: $\theta(r_c) = \pi/3$ and

$$q_-(r_c) = q_+(r_c) = q_c.$$

For $0 < r < r_c$, there are no positive roots.

For example, at $r = 0.65$ these exact expressions give

$$\begin{aligned} E_- &\approx (0.708427852,\ 1.413760738), \\ E_+ &\approx (0.896537216,\ 1.358028411). \end{aligned}$$

Together with $E_0 = (0, 2)$, these are all physical equilibria for this parameter value.

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5.6. **Equilibria and conditions for limit-cycle existence.** Consider

$$\begin{aligned} \dot{q} &= q\left[r + u - 2 - (r+0.1)(q-1)^2\right], \\ \dot{u} &= \varepsilon\left[2 - u + 2q(1-u)\right], \end{aligned} \qquad r, \varepsilon > 0. \tag{37}$$

Equilibria. Setting $\dot{q} = \dot{u} = 0$ gives the laminar equilibrium

$$E_0 = (0, 2).$$

Every other equilibrium has the form

$$E_* = \left(q_*, \frac{2(1+q_*)}{1+2q_*} \right), \tag{38}$$

where q_* is a real root of

$$2(r+0.1)q^3 - 3(r+0.1)q^2 + 2(1-r)q + 0.1 = 0. \tag{39}$$

Thus (39), together with (38), specifies all nonzero equilibria exactly.

There is always exactly one negative root $q_{\mathrm{n}} \in (-1/2, 0)$. Its equilibrium lies outside the physical region $q \geq 0$.

To count the positive roots, let q_c be the unique positive root of

$$44q_c^3 - 30q_c^2 - 3q_c - 1 = 0,$$

and define

$$r_c = \frac{1 + 22q_c}{10q_c(2-q_c)(1+2q_c)} - \frac{1}{10}.$$

Numerically,

$$q_c \approx 0.802140182, \qquad r_c \approx 0.645189835.$$

The physical equilibria are therefore:

Parameter range	Equilibria in $q \geq 0$
$0 < r < r_c$	E_0
$r = r_c$	$E_0,\ E_c$ (a double positive equilibrium)
$r > r_c$	$E_0,\ E_-,\ E_+$

where

$$E_\pm = \left(q_\pm, \frac{2(1+q_\pm)}{1+2q_\pm} \right), \qquad 0 < q_- < q_+ < 2.$$

The equilibrium coordinates do not depend on ε.

Linear stability. At E_0, the eigenvalues are

$$\lambda_1 = -0.1, \qquad \lambda_2 = -\varepsilon,$$

so the laminar equilibrium is always locally asymptotically stable.

At a nonzero equilibrium, the Jacobian is

$$J_* = \begin{pmatrix} 2(r+0.1)q_*(1-q_*) & q_* \\ -\dfrac{2\varepsilon}{1+2q_*} & -\varepsilon(1+2q_*) \end{pmatrix}. \tag{40}$$

Its trace and determinant are

$$T_* = 2(r+0.1)q_*(1-q_*) - \varepsilon(1+2q_*), \tag{41}$$

$$\Delta_* = 2\varepsilon q_* \left[(r+0.1)(q_*-1)(1+2q_*) + \frac{1}{1+2q_*} \right]. \tag{42}$$

The negative equilibrium and the lower positive equilibrium E_- are saddles. At the upper equilibrium E_+, one has $\Delta_+ > 0$; hence it is asymptotically stable when $T_+ < 0$ and unstable when $T_+ > 0$.

Test for a Hopf bifurcation. The relevant equilibrium for the birth of small limit cycles is E_+. The necessary spectral conditions are

$$T_+ = 0, \qquad \Delta_+ > 0.$$

The trace condition gives

$$\boxed{\varepsilon_H = \frac{2(r+0.1)q_+(1-q_+)}{1+2q_+}.} \tag{43}$$

This threshold is positive precisely when

$$\boxed{r_c < r < \frac{2}{3}.}$$

For each fixed r in this interval, $\Delta_+ > 0$ at $\varepsilon = \varepsilon_H$, and

$$\left.\frac{d\ \mathrm{Re}\,\lambda}{d\varepsilon}\right|_{\varepsilon=\varepsilon_H} = -\frac{1+2q_+}{2} \neq 0.$$

Thus the spectral and transversality conditions hold.

To conclude that a nondegenerate Hopf bifurcation actually produces a limit cycle, one must additionally verify that the first Lyapunov coefficient is nonzero. Its sign determines the stability of the cycle and the side of ε_H on which it exists. The condition $T_+ = 0$ alone is not sufficient.

For example, when $r = 0.65$,

$$q_+ \approx 0.896537216, \qquad u_+ \approx 1.358028411, \qquad \varepsilon_H \approx 0.049815126.$$

This is a concrete parameter point at which to perform the Hopf nondegeneracy test.
A direct nonexistence test. For the vector field (P, Q) in (37), the Dulac function $B(q, u) = 1/q$ on $q > 0$ gives

$$\frac{\partial(BP)}{\partial q} + \frac{\partial(BQ)}{\partial u} = 2(r+0.1)(1-q) - \frac{\varepsilon(1+2q)}{q}.$$

This expression is nonpositive throughout $q > 0$ if

$$\boxed{\varepsilon \geq (2-\sqrt{3})(r+0.1).}$$

The Bendixson–Dulac criterion then excludes all physical limit cycles. There are also no physical limit cycles for $r \leq r_c$.

Therefore, the Hopf test identifies where small cycles can arise, while the Dulac test excludes a region of parameter space. Neither test by itself determines the global number of limit cycles.

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6. Preliminary analysis of limit cycles

Throughout this section, let

$$A = r + \frac{1}{10}, \qquad h(q) = \frac{21}{10} - 2Aq + Aq^2, \tag{44}$$

and define the polynomials

$$\begin{aligned} H(q) &= (1+2q)h(q) - 2 - 2q \\ &= 2Aq^3 - 3Aq^2 + \left(\frac{11}{5} - 2A\right)q + \frac{1}{10}, \\ D(q) &= \varepsilon(1+2q) + qh'(q) \\ &= 2Aq^2 + 2(\varepsilon - A)q + \varepsilon. \end{aligned} \tag{45}$$

The original equations can then be written as

$$\dot{q} = q[u - h(q)], \qquad \dot{u} = \varepsilon[2 + 2q - (1+2q)u]. \tag{46}$$

6.1. **Invariant region and boundedness.** The half-plane $q \geq 0$ is positively invariant because $q = 0$ is an invariant line. Moreover, every solution with $q(0) > 0$ remains in $q > 0$ throughout its interval of existence.

The strip

$$\mathcal{S} = \{(q, u) : q \geq 0,\ 1 \leq u \leq 2\} \tag{47}$$

is positively invariant. Indeed,

$$\dot{u}|_{u=1} = \varepsilon > 0, \qquad \dot{u}|_{u=2} = -2\varepsilon q \leq 0.$$

More generally, if $q(0) \geq 0$, comparison at the horizontal boundaries gives

$$\min\{u(0), 1\} \leq u(t) \leq \max\{u(0), 2\}.$$

Since u is bounded and the leading term of $\dot{q}$ is $-Aq^3$, the component q also remains bounded. Consequently, every solution starting in $q \geq 0$ exists for all forward time.

Every nonconstant periodic orbit in the physical half-plane lies in

$$q > 0, \qquad 1 < u < 2. \tag{48}$$

To see this, first note that a periodic orbit cannot intersect the invariant line $q = 0$, on which $\dot{u} = \varepsilon(2 - u)$. At a maximum or minimum of u on a periodic orbit,

$$u = \frac{2(1+q)}{1+2q} = 1 + \frac{1}{1+2q},$$

which belongs to $(1, 2)$ when $q > 0$.

6.2. **Equilibria and their stability.** The laminar equilibrium

$$E_0 = (0, 2)$$

exists for all $r, \varepsilon > 0$. Its Jacobian is

$$J(E_0) = \begin{pmatrix} -\frac{1}{10} & 0 \\ -2\varepsilon & -\varepsilon \end{pmatrix},$$

so E_0 is locally asymptotically stable.

A positive equilibrium $E_* = (q_*, u_*)$ satisfies

$$u_* = \frac{2(1+q_*)}{1+2q_*}, \qquad H(q_*) = 0. \tag{49}$$

Equivalently, the positive-equilibrium branch is parameterized by

$$r = R(q) := \frac{1 + 22q}{10q(2-q)(1+2q)} - \frac{1}{10}, \qquad 0 < q < 2. \tag{50}$$

Differentiation yields

$$R'(q) = \frac{C(q)}{5q^2(q-2)^2(2q+1)^2}, \qquad C(q) = 44q^3 - 30q^2 - 3q - 1. \tag{51}$$

By Descartes' rule of signs, C has exactly one positive root. Denote it by q_{SN}. Since $C(0) < 0 < C(1)$, this root lies in $(0,1)$. Thus R decreases on $(0, q_{\mathrm{SN}})$ and increases on $(q_{\mathrm{SN}}, 2)$, with $R(q) \to +\infty$ at both endpoints.

Define

$$r_{\mathrm{SN}} = R(q_{\mathrm{SN}}). \tag{52}$$

Numerically,

$$q_{\mathrm{SN}} \approx 0.802140182, \qquad r_{\mathrm{SN}} \approx 0.645189835.$$

It follows that there are no positive equilibria for $r < r_{\mathrm{SN}}$, one double positive equilibrium for $r = r_{\mathrm{SN}}$, and two positive equilibria

$$E_{\pm} = \left(q_{\pm}, \frac{2(1+q_{\pm})}{1+2q_{\pm}} \right), \qquad q_- < q_+,$$

for $r > r_{\mathrm{SN}}$.

At a positive equilibrium, the Jacobian is

$$J_* = \begin{pmatrix} 2Aq_*(1-q_*) & q_* \\ -\dfrac{2\varepsilon}{1+2q_*} & -\varepsilon(1+2q_*) \end{pmatrix}. \tag{53}$$

In particular,

$$\operatorname{tr} J_* = -D(q_*), \qquad \det J_* = \varepsilon q_* H'(q_*). \tag{54}$$

For $r > r_{\mathrm{SN}}$,

$$H'(q_-) < 0, \qquad H'(q_+) > 0.$$

Hence E_- is a saddle, whereas E_+ is asymptotically stable when $D(q_+) > 0$ and unstable when $D(q_+) < 0$.

Since $R(1) = 2/3$, the upper equilibrium satisfies

$$q_+ < 1 \quad \Longleftrightarrow \quad r_{\mathrm{SN}} < r < \frac{2}{3}.$$

For $r \geq 2/3$, one has $q_+ \geq 1$ and therefore $D(q_+) > 0$ for every $\varepsilon > 0$.

6.3. **Hopf bifurcation and local limit cycles.** A positive Hopf threshold exists precisely on the upper branch with $r_{\mathrm{SN}} < r < 2/3$. It is given by

$$\varepsilon_H(r) = \frac{2Aq_+(1-q_+)}{1+2q_+}. \tag{55}$$

At this threshold,

$$D(q_+) = 0, \qquad \omega^2 := \varepsilon_H q_+ H'(q_+) > 0.$$

The equilibrium coordinates are independent of ε, and

$$\left. \frac{\partial \operatorname{Re} \lambda}{\partial \varepsilon} \right|_{\varepsilon = \varepsilon_H} = -\frac{1+2q_+}{2} < 0. \tag{56}$$

Thus the transversality condition holds.

For the calculation of the Hopf coefficient, the polynomial substitution

$$v = u - h(q)$$

gives

$$\dot{q} = qv, \qquad \dot{v} = -D(q)v - \varepsilon H(q). \tag{57}$$

This auxiliary system remains cubic polynomial and uses the original time variable.

At the Hopf point, set $s = q_+$ and introduce the linear coordinates

$$X = q - s, \qquad Y = -\frac{s}{\omega} v.$$

The linear part becomes $\dot{X} = -\omega Y$, $\dot{Y} = \omega X$. The cubic coefficient in the corresponding radial normal form is

$$\ell = \frac{D'(s)H''(s) - D''(s)H'(s)}{16H'(s)}, \tag{58}$$

where D is evaluated at $\varepsilon = \varepsilon_H$. Using $H(s) = 0$ and (55), this simplifies to

$$\ell = \frac{(22s+1)\left(88s^4 + 176s^3 - 186s^2 + 64s + 5\right)}{80s(2-s)(2s+1)^2 C(s)}. \tag{59}$$

For $q_{\mathrm{SN}} < s < 1$, the denominator is positive. Moreover,

$$88s^4 + 176s^3 - 186s^2 + 64s + 5 = 88s^4 + s(176s^2 - 186s + 64) + 5 > 0,$$

because the quadratic $176s^2 - 186s + 64$ has positive leading coefficient and negative discriminant. Consequently,

$$\ell > 0. \tag{60}$$

Therefore, for every fixed $r \in (r_{\mathrm{SN}}, 2/3)$, the Hopf bifurcation is nondegenerate and subcritical. A unique small-amplitude unstable limit cycle bifurcates from E_+ for $\varepsilon > \varepsilon_H(r)$ sufficiently close to $\varepsilon_H(r)$. The equilibrium inside this cycle is asymptotically stable.

This uniqueness is local: it concerns cycles sufficiently close to the Hopf equilibrium and does not exclude additional cycles elsewhere in the phase plane.

6.4. **Restrictions on the location of periodic orbits.** At an extremum of q on a nonconstant periodic orbit, $\dot{q} = 0$ and hence $u = h(q)$. Differentiating the first equation of (46) gives, at such a point,

$$\ddot{q} = -\varepsilon q H(q). \tag{61}$$

The extremum cannot satisfy $H(q) = 0$, since it would then be an equilibrium point.

If $r \le r_{\mathrm{SN}}$, one has $H(q) \ge 0$ for all $q > 0$. Equation (61) then excludes the minimum required by any nonconstant periodic orbit. Thus

$$0 < r \le r_{\mathrm{SN}} \quad \Longrightarrow \quad \text{no limit cycles in } q \ge 0. \tag{62}$$

Suppose now that $r > r_{\mathrm{SN}}$. Since H is negative precisely between its positive roots, the minimum and maximum of q along any periodic orbit satisfy

$$q_- < q_{\min} < q_+ < q_{\max} < 2. \tag{63}$$

For the final inequality, recall that $u < 2$ along the orbit, whereas

$$h(q) = \frac{21}{10} + Aq(q-2) \ge \frac{21}{10} \qquad \text{for } q \ge 2.$$

Hence a q-maximum cannot occur at $q \ge 2$.

In particular, every physical limit cycle excludes the saddle E_- and the boundary equilibrium E_0. By the index theorem for planar periodic orbits, it surrounds E_+.

6.5. **A Bendixson–Dulac exclusion region.** Let P and Q denote the right-hand sides of (2). On the simply connected half-plane $q > 0$, choose the Dulac function

$$B(q,u) = \frac{1}{q}.$$

A direct calculation gives

$$\frac{\partial(BP)}{\partial q} + \frac{\partial(BQ)}{\partial u} = 2A(1-q) - \frac{\varepsilon(1+2q)}{q} = -\frac{D(q)}{q}. \tag{64}$$

The quadratic polynomial D is nonnegative for all $q > 0$ if

$$\varepsilon \ge (2 - \sqrt{3})A. \tag{65}$$

Indeed, for $\varepsilon \geq A$ all its coefficients are nonnegative. For $0 < \varepsilon < A$, its minimum is

$$\min_{q>0} D(q) = \varepsilon - \frac{(A-\varepsilon)^2}{2A},$$

which is nonnegative precisely when $\varepsilon \geq (2-\sqrt{3})A$.

The Bendixson–Dulac criterion therefore yields

$$\boxed{\varepsilon \geq (2-\sqrt{3})\left(r + \frac{1}{10}\right) \quad \Longrightarrow \quad \text{no limit cycles in } q \geq 0.} \tag{66}$$

At equality, the Dulac divergence vanishes only on one vertical line, which does not affect the conclusion.

6.6. **Consequences for the global cycle problem.** The preceding results restrict the search for physical limit cycles to

$$r > r_{\mathrm{SN}}, \qquad 0 < \varepsilon < (2-\sqrt{3})\left(r + \frac{1}{10}\right).$$

Every such cycle surrounds the upper positive equilibrium and remains in the region specified by (48) and (63). Furthermore, a small unstable cycle is known to exist on the subcritical side of the Hopf curve (55).

These conclusions establish local existence and explicit global exclusion regions. They do not yet determine the total number of limit cycles or the global continuation of the Hopf branch. In particular, uniqueness away from the Hopf point and possible termination in a saddle homoclinic loop require additional analysis.

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